Stat_Project

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(knitr)
library(hoopR)
library(rvest)
```

```
##
## Attaching package: 'rvest'
##
## The following object is masked from 'package:readr':
##
##     guess_encoding
```

```r
library(stringr)
library(stringi)

knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)
```

```r
rawgamelogs <- load_nba_player_box(seasons = 2025)

library(dplyr)

team_totals <- rawgamelogs %>%
  group_by(game_id, team_id) %>%
  summarise(
    team_FGA = sum(field_goals_attempted, na.rm = TRUE),
    team_FTA = sum(free_throws_attempted, na.rm = TRUE),
    team_TOV = sum(turnovers, na.rm = TRUE),
    team_minutes = sum(minutes, na.rm = TRUE)
  )

rawgamelogs2 <- rawgamelogs %>%
  left_join(team_totals, by = c("game_id", "team_id"))

rawgamelogs2 <- rawgamelogs2 %>%
  mutate(
    USG = 100 *
```

```r
      (field_goals_attempted + 0.44 * free_throws_attempted + turnovers) *
      team_minutes /
      ((team_FGA + 0.44 * team_FTA + team_TOV) * 5 * minutes)
  )

rawgamelogs2 <- rawgamelogs2 %>%
  filter(minutes > 0) %>%
  filter(USG < 100)
```

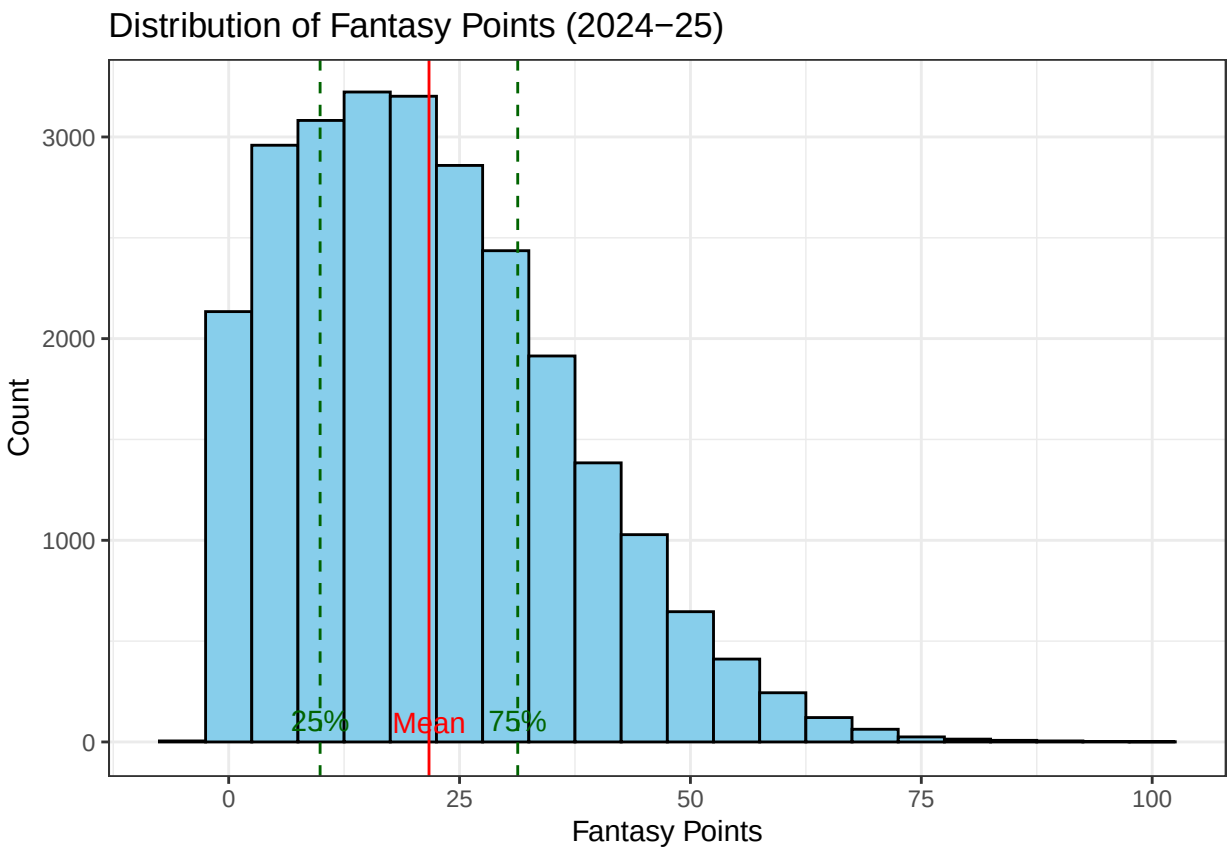
```r
plot_data <- GameLogs %>%
  filter(
    !is.na(FantasyPoints),
    !is.na(Minutes),
    !is.na(Points),
    !is.na(Rebounds),
    !is.na(Assists),
    !is.na(USG)
  )
```

```r
methods_table <- tibble(
  Variable = c("FantasyPoints", "Minutes", "Points", "Assists", "Rebounds", "Position"),
  Meaning = c(
    "Overall player performance score",
    "Minutes played in the game",
    "Points scored in the game",
    "Number of assists",
    "Number of rebounds",
    "Player position"
  ),
  Units = c(
    "Points",
    "Minutes",
    "Points",
    "Count",
    "Count",
    "Category"
  ),
  Min = c(
    min(plot_data$FantasyPoints, na.rm = TRUE),
    min(plot_data$Minutes, na.rm = TRUE),
    min(plot_data$Points, na.rm = TRUE),
    min(plot_data$Assists, na.rm = TRUE),
    min(plot_data$Rebounds, na.rm = TRUE),
    NA
  ),
  Max = c(
    max(plot_data$FantasyPoints, na.rm = TRUE),
    max(plot_data$Minutes, na.rm = TRUE),
    max(plot_data$Points, na.rm = TRUE),
    max(plot_data$Assists, na.rm = TRUE),
    max(plot_data$Rebounds, na.rm = TRUE),
    NA
  ),
  Median = c(
```

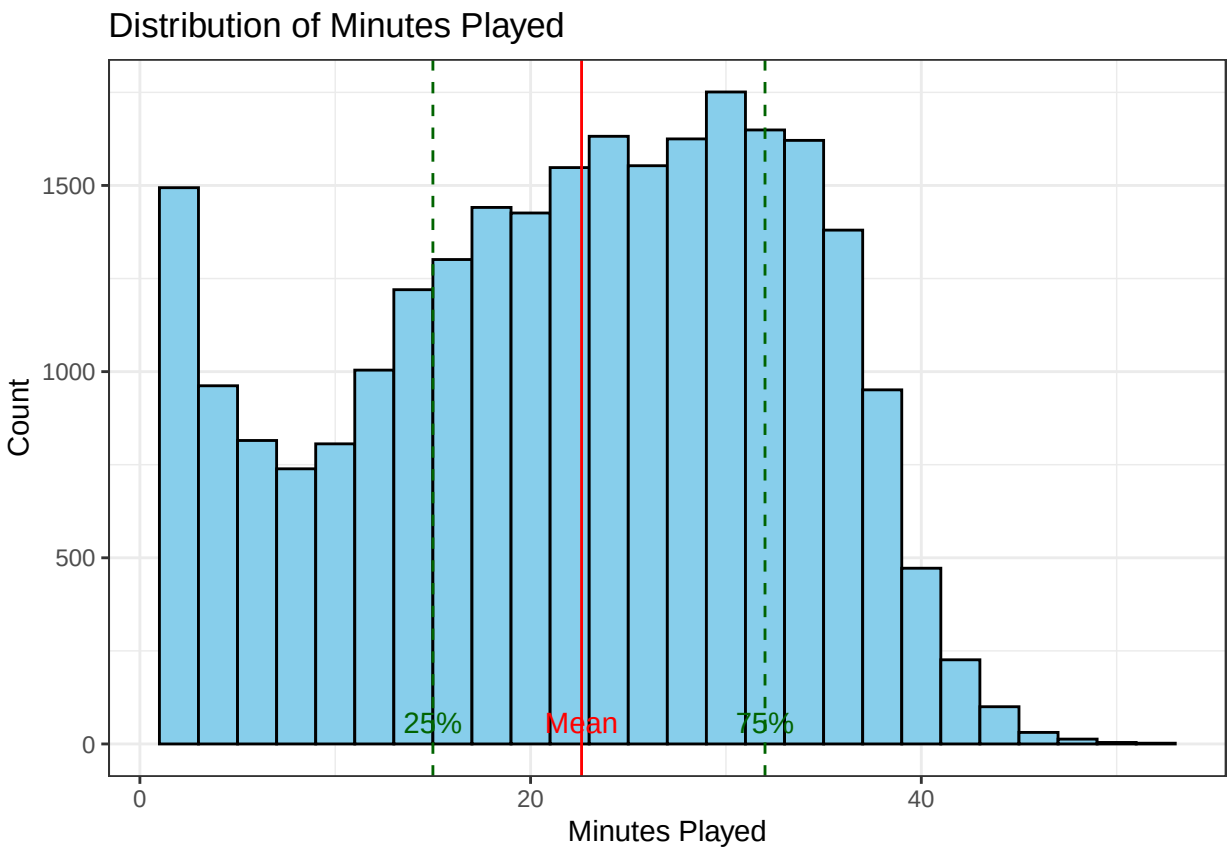
```r
    median(plot_data$FantasyPoints, na.rm = TRUE),
    median(plot_data$Minutes, na.rm = TRUE),
    median(plot_data$Points, na.rm = TRUE),
    median(plot_data$Assists, na.rm = TRUE),
    median(plot_data$Rebounds, na.rm = TRUE),
    NA
  ),
  Std = c(
    sd(plot_data$FantasyPoints, na.rm = TRUE),
    sd(plot_data$Minutes, na.rm = TRUE),
    sd(plot_data$Points, na.rm = TRUE),
    sd(plot_data$Assists, na.rm = TRUE),
    sd(plot_data$Rebounds, na.rm = TRUE),
    NA
  )
)

methods_table
```

```
## # A tibble: 6 x 7
##   Variable      Meaning                          Units    Min   Max Median   Std
##   <chr>         <chr>                            <chr>  <dbl> <dbl>  <dbl> <dbl>
## 1 FantasyPoints Overall player performance score Points    -3  98.3   19.9 15.0
## 2 Minutes       Minutes played in the game       Minut~     1  53     24   11.0
## 3 Points        Points scored in the game        Points     0  61      9    8.86
## 4 Assists       Number of assists                Count      0  22      2    2.61
## 5 Rebounds      Number of rebounds               Count      0  28      3    3.47
## 6 Position      Player position                  Categ~    NA  NA     NA   NA
```

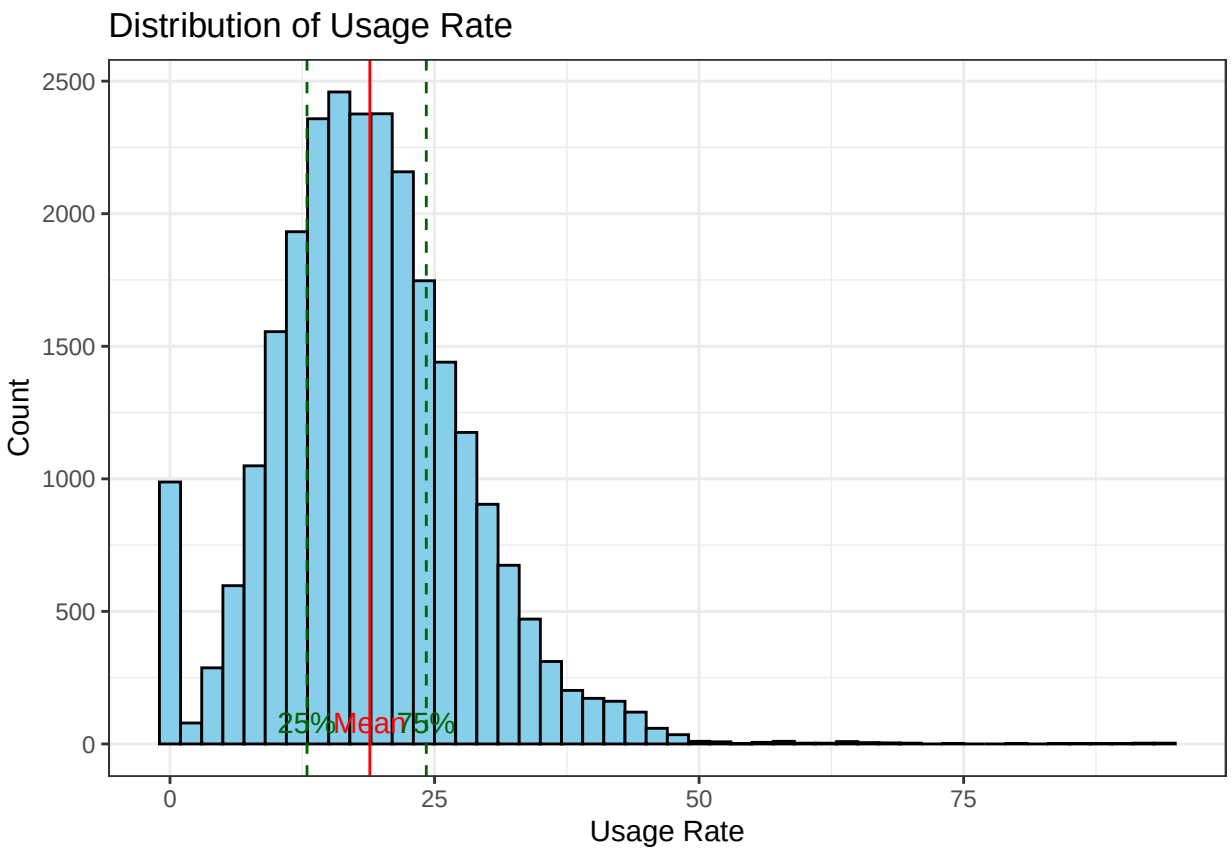
```r
mean_fp <- mean(plot_data$FantasyPoints, na.rm = TRUE)
q25 <- quantile(plot_data$FantasyPoints, 0.25, na.rm = TRUE)
q75 <- quantile(plot_data$FantasyPoints, 0.75, na.rm = TRUE)
ggplot(plot_data, aes(x = FantasyPoints)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  geom_vline(xintercept = mean_fp, color = "red", linewidth = 0.5) +
  annotate("text", x = mean_fp, y = -10, label = "Mean", color = "red", angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q25, color = "darkgreen", linetype = "dashed") +
  annotate("text", x = q25, y = 0, label = "25%", color = "darkgreen", angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q75, color = "darkgreen", linetype = "dashed") +
  annotate("text", x = q75, y = 0, label = "75%", color = "darkgreen", angle = 0, vjust = -0.5) +
  labs(
    title = "Distribution of Fantasy Points (2024-25)",
    x = "Fantasy Points",
    y = "Count"
  ) +
  theme_bw()
```



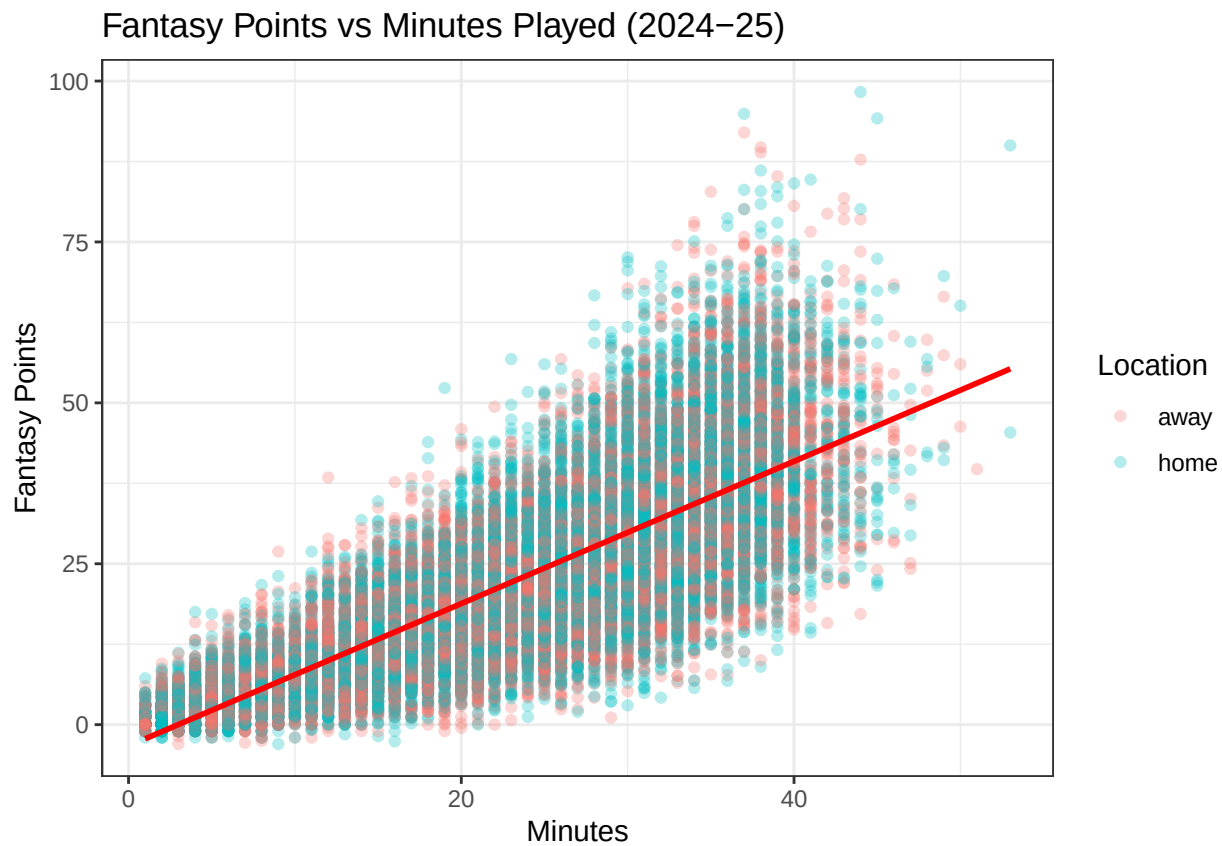

```
mean_min <- mean(plot_data$Minutes, na.rm = TRUE)
q_25_min <- quantile(plot_data$Minutes, 0.25, na.rm = TRUE)
q_75_min <- quantile(plot_data$Minutes, 0.75, na.rm = TRUE)
ggplot(plot_data, aes(x = Minutes)) +
  geom_histogram(binwidth = 2, fill = 'skyblue', color = 'black')+
  geom_vline(xintercept = mean_min, color = 'red', linewidth = 0.5) +
  annotate('text', x = mean_min, y = 0, label = 'Mean', color = 'red', angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q_25_min, color = 'darkgreen', linetype = 'dashed') +
  annotate('text', x = q_25_min, y = 0, label = '25%', color = 'darkgreen', angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q_75_min, color = 'darkgreen', linetype = 'dashed') +
  annotate('text', x = q_75_min, y = 0, label = '75%', color = 'darkgreen', angle = 0, vjust= -0.5) +
  labs(title = 'Distribution of Minutes Played',
       x = 'Minutes Played',
       y = 'Count') +
  theme_bw()
```



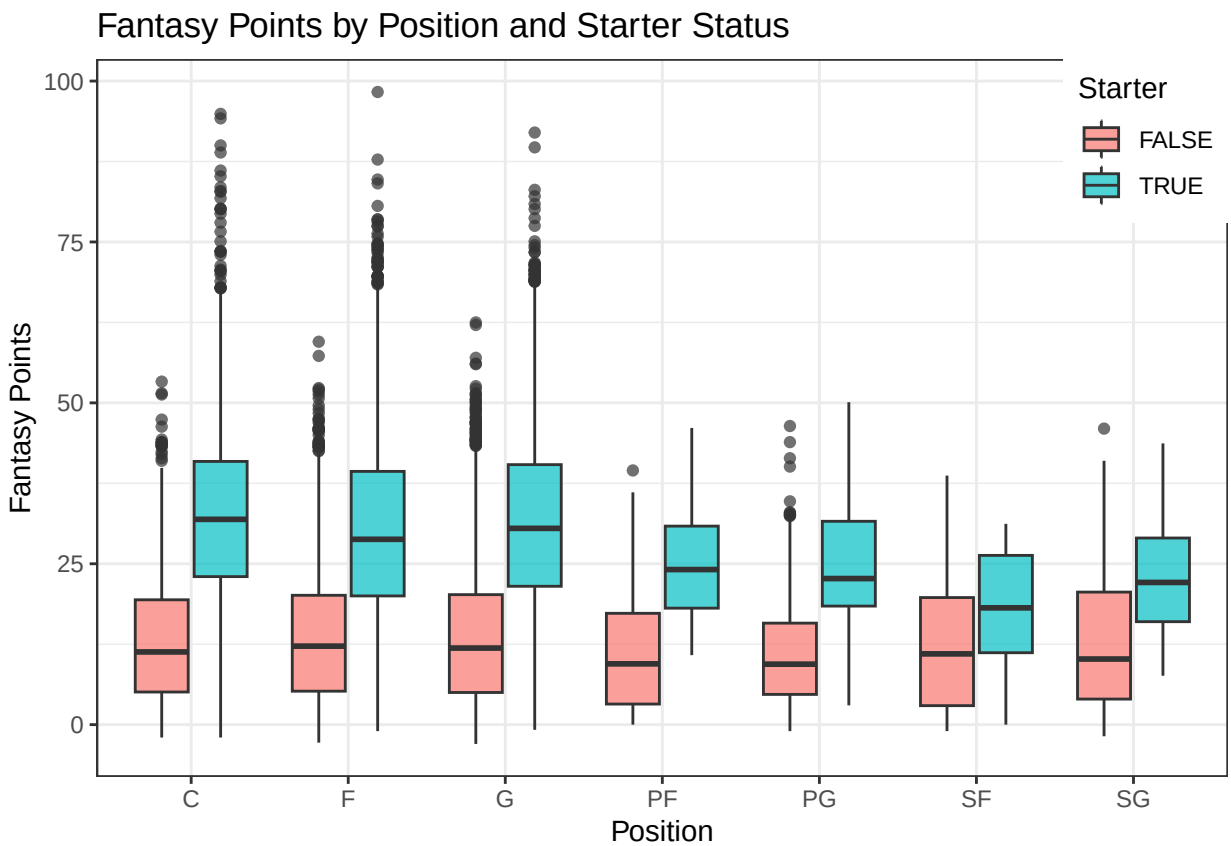

```
mean_usg <- mean(plot_data$USG, na.rm = TRUE)
q_25_usg <- quantile(plot_data$USG, 0.25, na.rm = TRUE)
q_75_usg <- quantile(plot_data$USG, 0.75, na.rm = TRUE)
ggplot(plot_data, aes(x = USG)) +
  geom_histogram(binwidth = 2, fill = 'skyblue', color = 'black')+
  geom_vline(xintercept = mean_usg, color = 'red', linewidth = 0.5) +
  annotate('text', x = mean_usg, y = 0, label = 'Mean', color = 'red', angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q_25_usg, color = 'darkgreen', linetype = 'dashed') +
  annotate('text', x = q_25_usg, y = 0, label = '25%', color = 'darkgreen', angle = 0, vjust = -0.5) +
  geom_vline(xintercept = q_75_usg, color = 'darkgreen', linetype = 'dashed') +
  annotate('text', x = q_75_usg, y = 0, label = '75%', color = 'darkgreen', angle = 0, vjust= -0.5) +
  labs(title = 'Distribution of Usage Rate',
       x = 'Usage Rate',
       y = 'Count') +
  theme_bw()
```



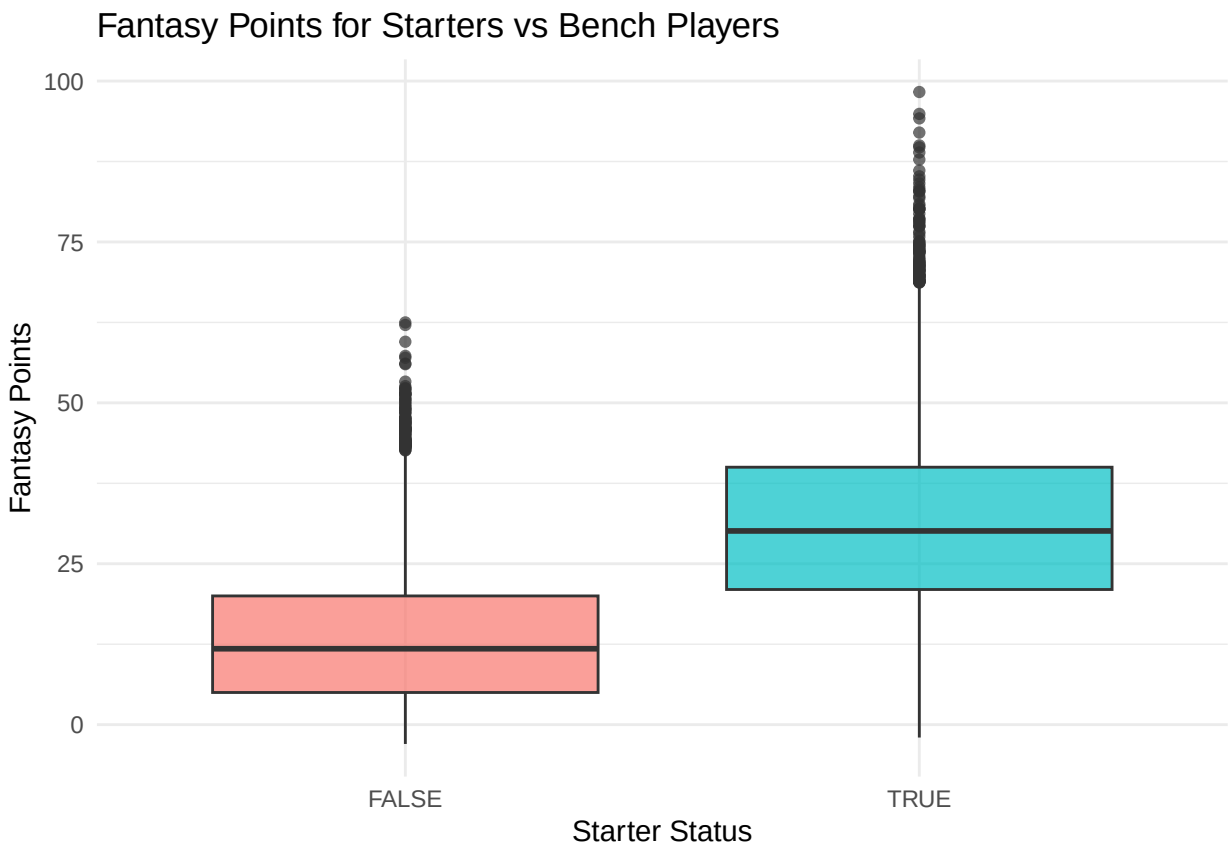

```r
ggplot(plot_data, aes(x = Minutes, y = FantasyPoints, color = Location)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Fantasy Points vs Minutes Played (2024-25)",
    x = "Minutes",
    y = "Fantasy Points"
  ) +
  theme_bw()
```




```r
ggplot(plot_data, aes(x = Position, y = FantasyPoints, fill = Starter)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    title = "Fantasy Points by Position and Starter Status",
    x = "Position",
    y = "Fantasy Points",
    fill = "Starter"
  ) +
  theme_bw() +
  theme(
    legend.position = c(1, 1),
    legend.justification = c(1, 1)
  )
```




```r
ggplot(plot_data, aes(x = Starter, y = FantasyPoints, fill = Starter)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    title = "Fantasy Points for Starters vs Bench Players",
    x = "Starter Status",
    y = "Fantasy Points"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Fantasy Points for Starters vs Bench Players
100
75
50
25
0
Fantasy Points
FALSE
TRUE
Starter Status